

Introduction to LLM Explainability

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CSCI 444 Guest Lecture

23 March 2026

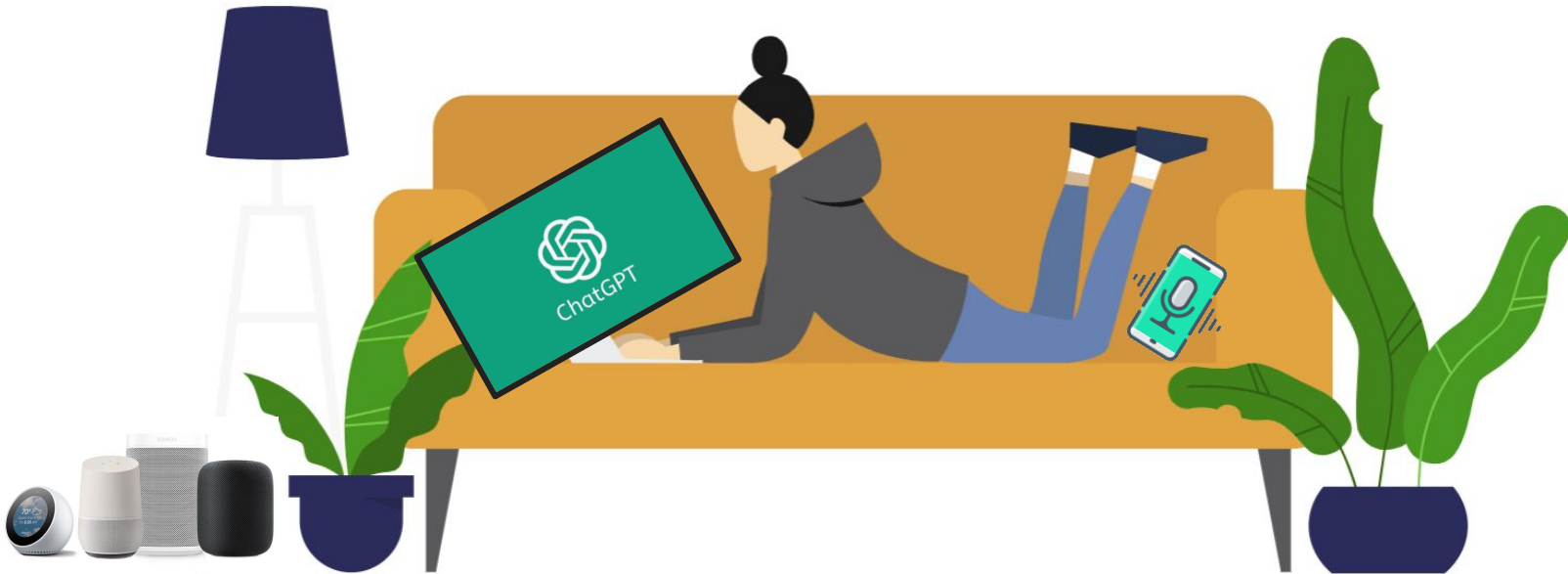
About me!

- 5th year PhD candidate at USC working with Xiang Ren and Swabha Swayamdipta
- Primarily interested in: Humans x AI.
 - How can AI systems be made better for people?
 - Personalization, Utility, Interaction, Reducing cognitive burden
- Spent time doing research with Apple and Amazon
- Fun stuff: I play 🎤🎹🎸🥁💻, read a lot of romance fiction, and making high protein variants of my favourite food!
- In my previous life (pre-PhD), I worked for an investment bank that made me join this PhD program!

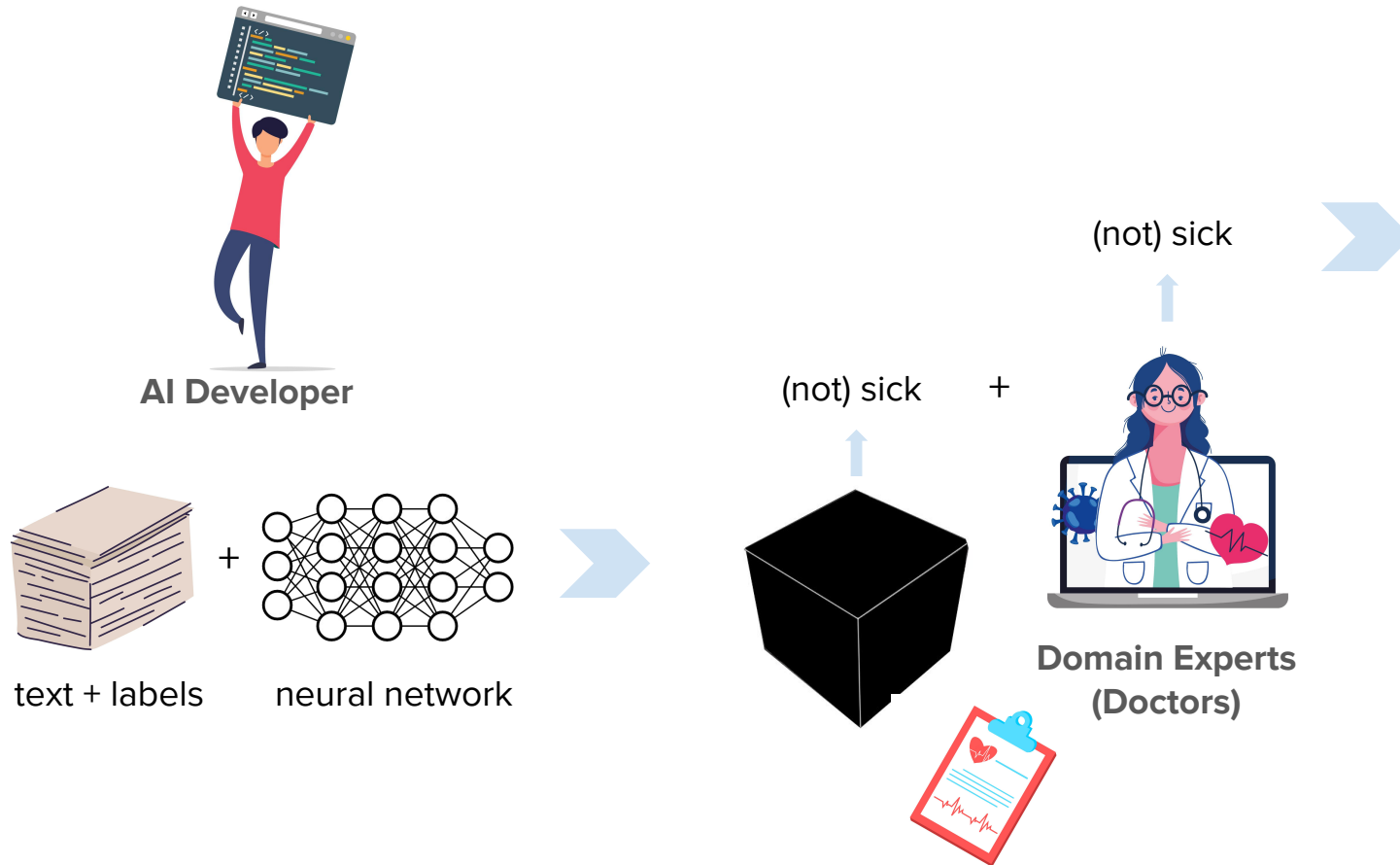


Note: I love mentoring! I am hiring students to work on exciting projects! Previously, mentored multiple CURVE and MS/visiting students!

ML/AI technology has become an integral part
of most people's daily lives



People Affected by AI (Patients)



Increasingly harder to opt out



Doctors

Promised Benefits

- Faster diagnosis
- Better treatment
- Less burnout & stress

Risks

- Hurt their patients
- Bad performance review
- Getting fired
- Lawsuits

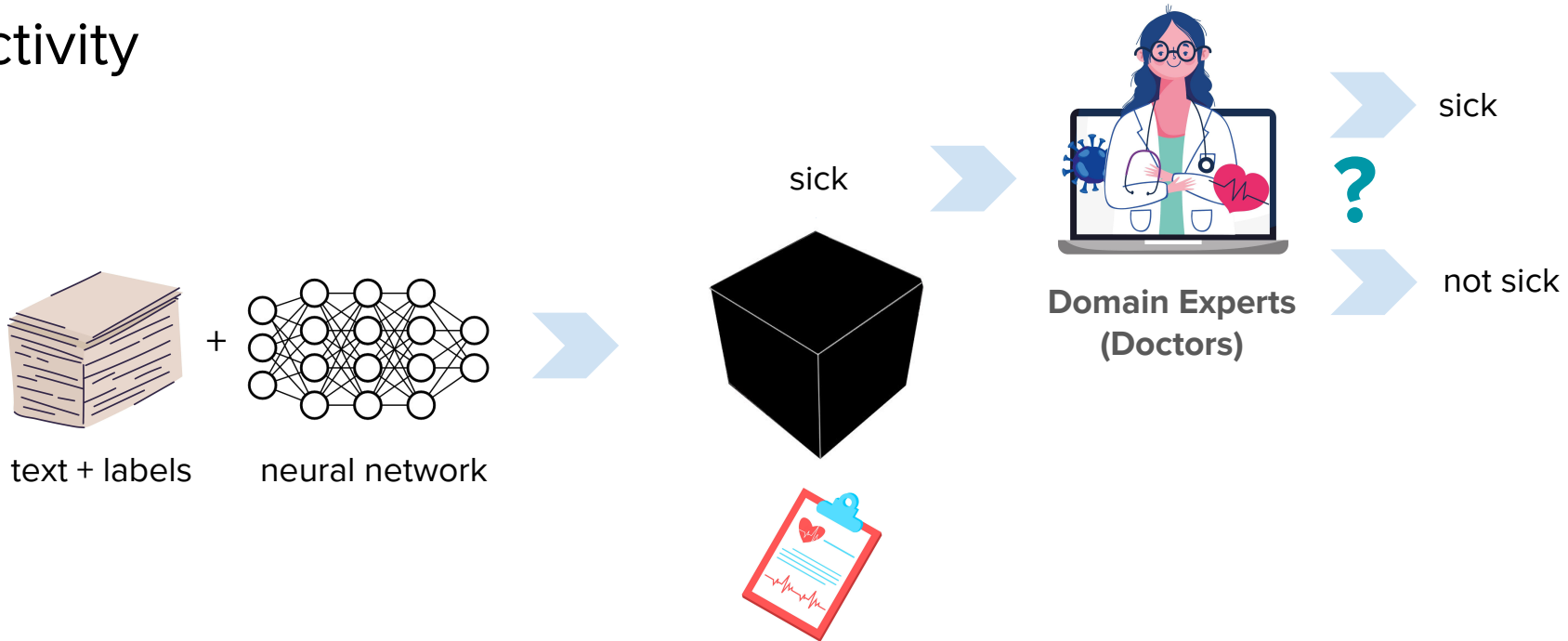


Patients

- Faster diagnosis
- Better treatment

- Unintuitive care
- Possibly Wrong treatment
- Death

Activity



Imagine you are a *doctor* that is using an LLM to make a medical diagnosis. However, the LLM is seeming to contradict your opinion.
Task: Design a tiny experiment in which you would test whether the model is making a *correct* decision.

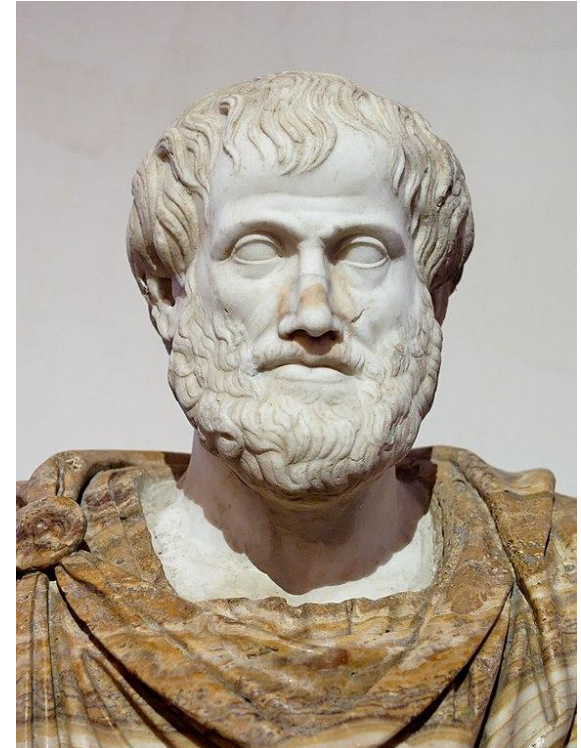
This Lecture

A brief primer on explaining model decisions, mostly taken from research work in this community in the past ~5 years!

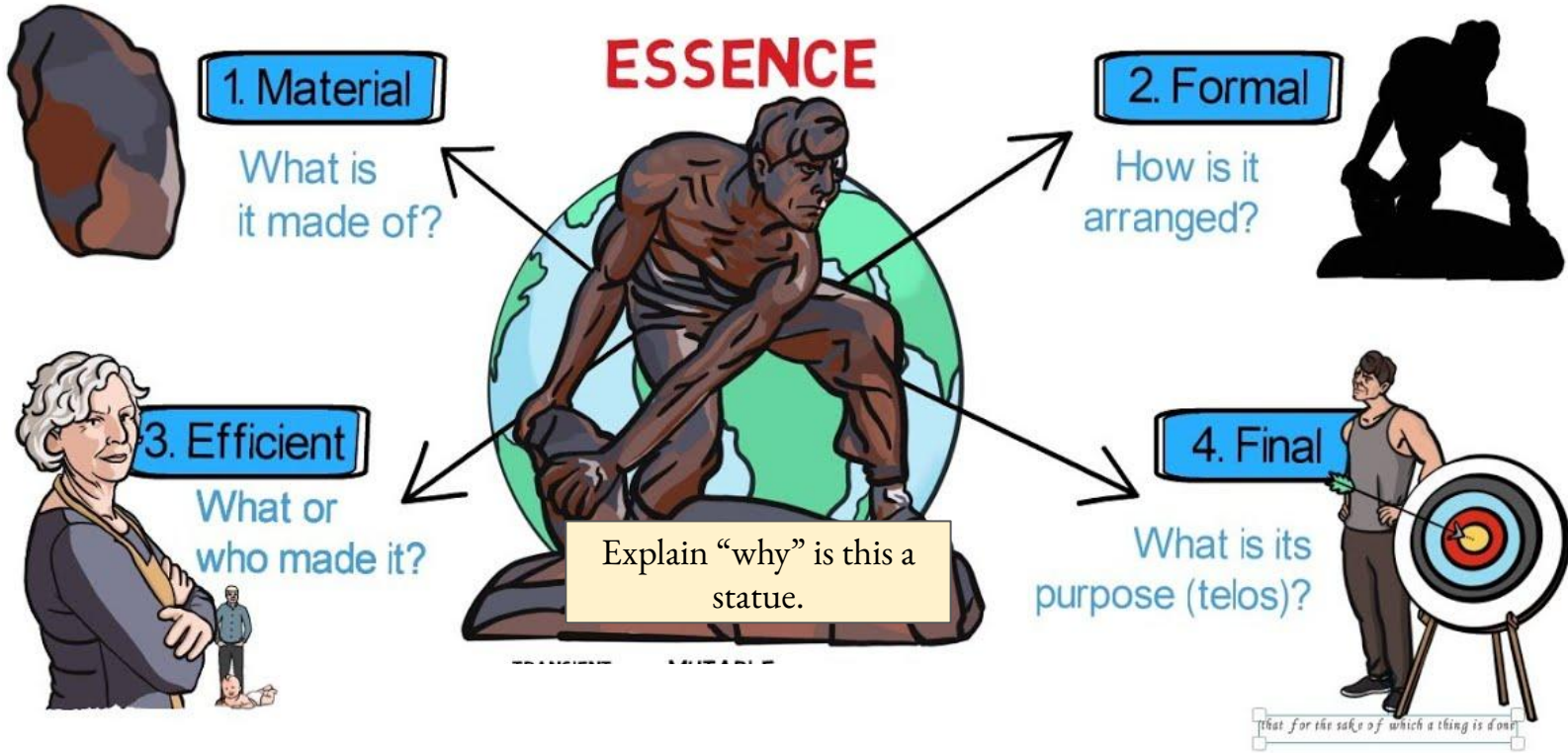
- A brief history of how (and why) we think about explanations
- A short primer on different cutting edge methods for explaining model decisions:
 - Uncertainty Estimation
 - Natural Language Explanations
 - Highlighting spans
 - Inherently Interpretable models
 - Token Attribution
 - Training Data
 - Contrastive Editing
 - Modern Interpretability: Activation Patching
- How can we evaluate explanations?

Let's dial back to...Aristotle 🧔

- First (known) thought experiment on “why” certain things occur.
- **Coined** - Explanations are answers to “why” questions
- Defined **four** primary ways to answer “why” questions → thus creating the first taxonomy of explanations



ARISTOTLE'S FOUR CAUSES





Why is this input assigned this answer?

- *Which input words are important?*
- *Is it because of certain interaction between words?*
- *Or some human-interpretable concept?*
- *Maybe some training examples?*
- *Just tell me why in plain English. :)*



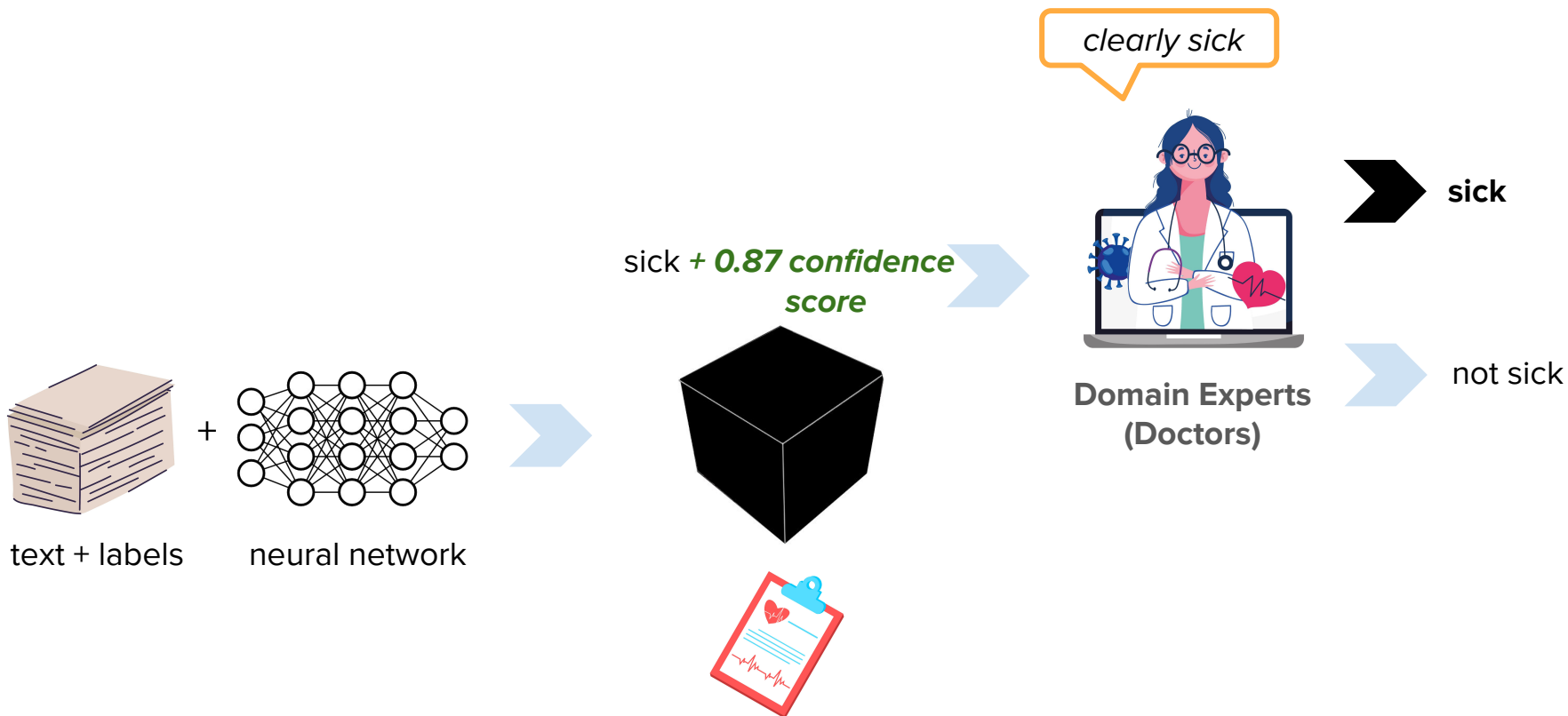
Topic 1

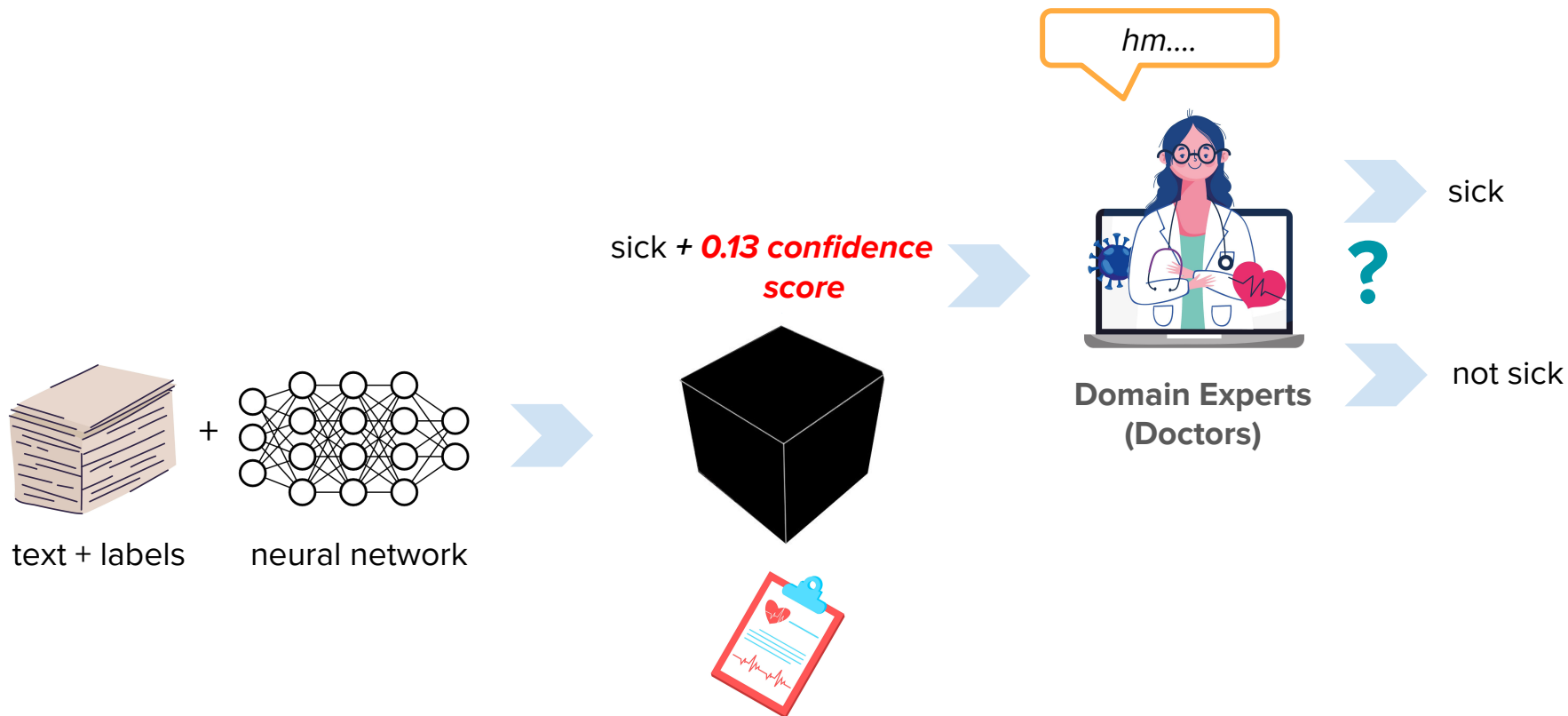
Why is this input assigned this answer?



Uncertainty Estimation

How confident is the model in its decision?





Uncertainty sources

Epistemic (source: modeling) uncertainty

- Model assumptions are wrong
 - ◆ e.g; Bag-Of-Word models do not model word order
- Lack training data
- This type of uncertainty can be reduced by collecting more data and/or improving the model

Aleatoric (source: data) uncertainty

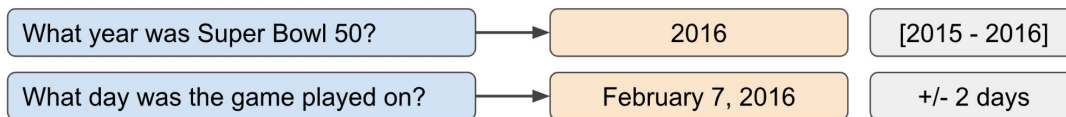
- The uncertainty is inherent in the task data itself
 - ◆ People disagree about what should or should not be labelled as hateful
- Collecting more data or changing the model won't reduce the uncertainty

Humans express uncertainty in various ways

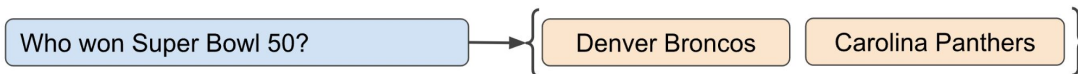
- As a **numerical value** (e.g. in $[0,1]$) returned with each prediction



- As a **confidence interval** around a numerical value:



- A **set of candidate answers**:



- As a **decision to abstain from answer**:



Obtaining uncertainty estimates from LLMs - I

→ Softmax-based measures

- ◆ Entropy of the softmax scores
- ◆ The maximum value

→ “Self”-estimation

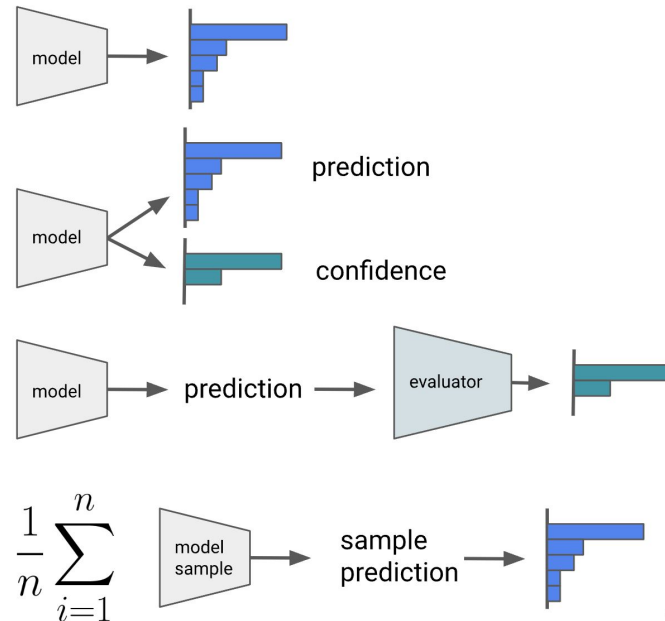
- ◆ Model predicts its own confidence scores

→ Separate independent evaluator

- ◆ A separate model evaluates the prediction

→ Model-inherent measures

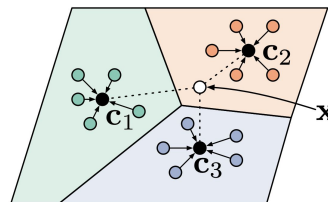
- ◆ Bayesian modeling
- ◆ Sampling-based estimates



Obtaining uncertainty estimates from LLMs - II

→ Distance in some space

- ◆ Average k-NN distance from the training data
- ◆ Outlier detection based on density estimates



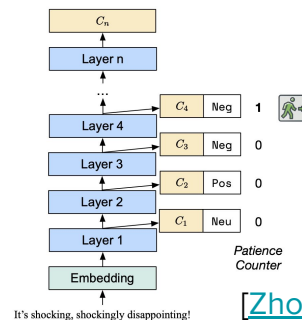
[Snell et al., 2017]

→ Consistency across label-preserving augmentations



→ Other model behaviors that can serve as proxies

- ◆ Repeated similar predictions by intermediate layers



[Zhou et al., 2020]

Topic 2

Why is this input assigned this answer?

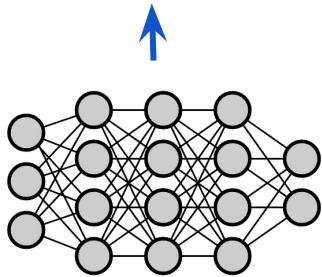


Free-text explanations

Chain-of-Thoughts

In plain English, why is this input
assigned this label?

Misleading because not every American over 65 can get these cards since they are not provided by Medicare, the federal health insurance program for senior citizens. They are offered as a benefit to some customers by private insurance companies that sell Medicare Advantage plans. The cards are available in limited geographic areas. Only the chronically ill qualify to use the cards for items such as food and produce.



+ documents from
the Web

4 January at 10:55 · 🌐

I AM SO HAPPY I JUST LEARNED THIS!
As an American over 65, I qualified for the "Elderly Spend Card", which pays for my groceries, my dental, and my prescription refills. All I did to qualify, was tap the image below, entered my zip and I got my flex card in the mail a week later!



Seniors Must Claim Today:->

453457 People Already Register

Learn More

👍 1.6K

1.2K comments 318

shares

👍 Like

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➦ Share

Self-explaining with **free-text explanations**:

Given in **plain language**, immediately provide the **gist of why** is the input labeled as it is





[Google I/O 2022](#)

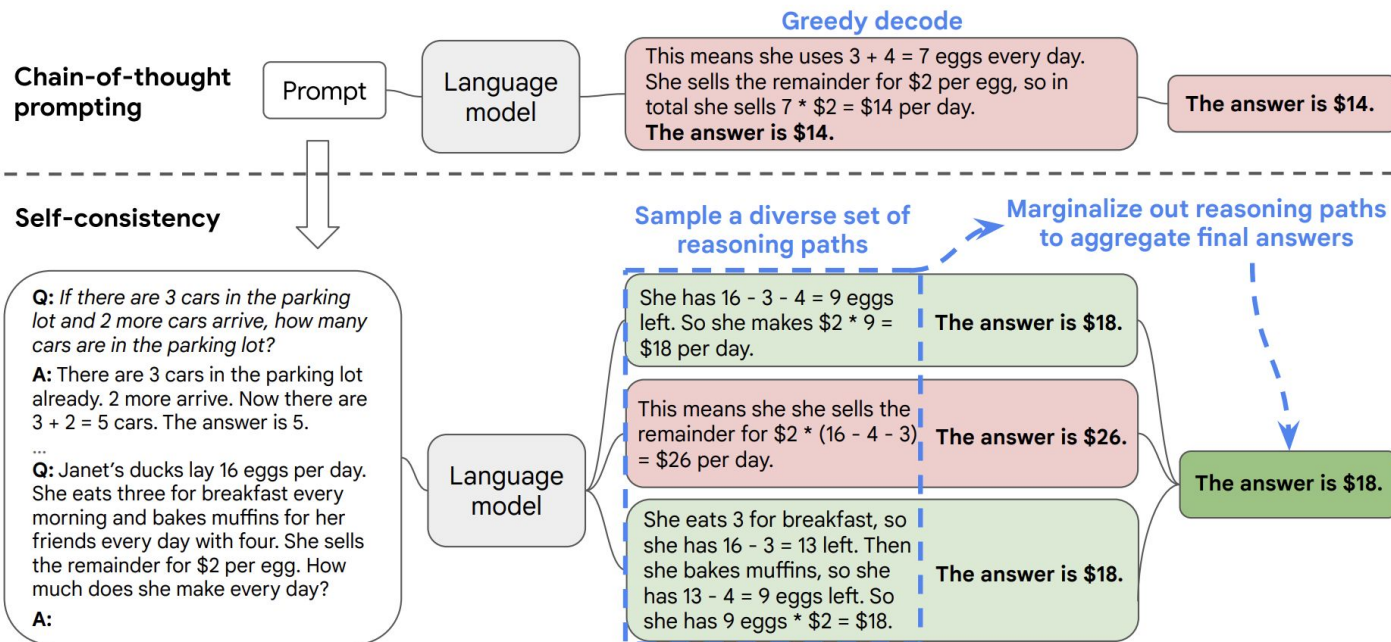
Chain-of-thought prompting

This approach:

- prompt input = {task_instruction}{labeled example #1 as a QA instance with an **explanation**}...{labeled example #n as a QA instance with an **explanation**} {eval example input} **Answer: Let's think step by step.**
- prompt output = {**explanation**} **So the answer is** {answer}.
- no further training (changing of model weights)

is called **chain-of-thought prompting**

Self-Consistency



How to train a model to generate label+free-text explanations?

Step 1:

Find some human-written explanations[◇]

Step 2:

Finetune a pretrained transformer-based generation models (**AutoModelForCausalLM**) to generate a sequence **<label> because <explanation>** [or whatever works best for your model!] conditioned on a given input:

- A sequence2sequence (seq2seq) approach [[Sutskever et al., 2014](#)]
- Large models (text generation improves with model size)
- Models that are instruction-finetuned with CoTs like modern reasoning models are “primed” to generate these things
- Models pretrained only with a MLM objective don’t work well here (BERT, RoBERTa, etc.)

[◇][Wiegreffe* and Marasovic*, 2021] ⇒ <https://exnlpdatasets.github.io/>

Example Free-text Explanation

Question: Where is a frisbee in play likely to be?

Answer choices: outside, park, roof, tree, air

Free-text explanation: A frisbee is a concave plastic disc designed for skimming through the air as an outdoor game so while in play it is most likely to be in the air.

[Aggarwal et al., 2021]

Topic 3

Why is this input assigned this answer?



highlighting

Which parts of the input are responsible for this prediction?

Highlighting methods highlight input features (pixels, words, etc.) that were important for a model prediction

Fun Fact! Humans actually do something pretty similar, when we are skimming large blobs of text!

Input highlights are also known as:

1. Saliency maps (for images)
2. Sensitivity maps (for images)
3. Input (feature) attribution
4. Input feature importance
5. Input feature relevance
6. Input feature contribution
7. Extractive rationales

Highlighting methods **highlight input features** (pixels, words, etc.) that were **important** for a model **prediction**

Fun Fact! **Humans** actually do something **pretty similar**, when we are **skimming large blobs** of text!

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Highlighting Techniques in General

- Compute the relative “*importance*” of each feature (e.g., word) in the input
- “Importance” is, loosely:
if you change or remove the feature, how much is the prediction affected?

Example of highlighting: Masked language modeling

Sentence

The doctor ran to the emergency room to see [MASK] patient.

Mask 1

Prediction	Score
The doctor ran to the emergency room to see his patient .	38.3%
The doctor ran to the emergency room to see the patient .	36.9%
The doctor ran to the emergency room to see another patient .	8.1%
The doctor ran to the emergency room to see a patient .	7.3%
The doctor ran to the emergency room to see her patient .	6%

Example of highlighting: Masked language modeling

Prediction

Score

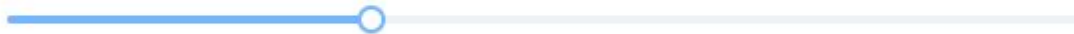
The doctor ran to the emergency room to see his patient .



38.3%

SENTENCE

[CLS] The doctor ran to the emergency room to see [MASK] patient . [SEP]



Visualizing the top 2 most important words.

Example of highlighting: Sentiment classification

Sentence

a very well-made, funny and entertaining picture.

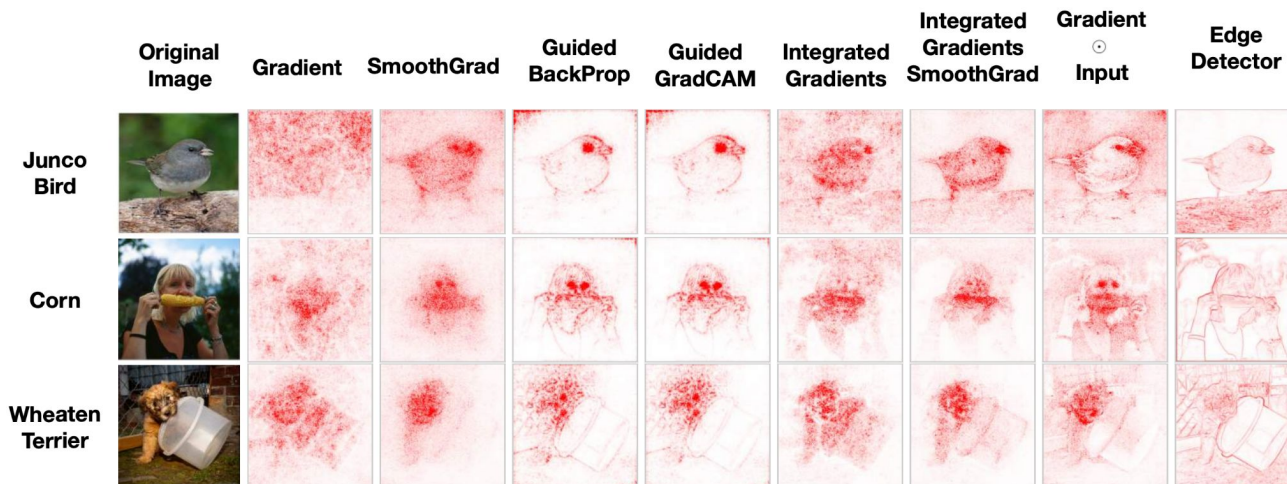
The model is **very confident** that the sentence has a **positive** sentiment.

SENTENCE

a very well - made , funny and entertaining picture .

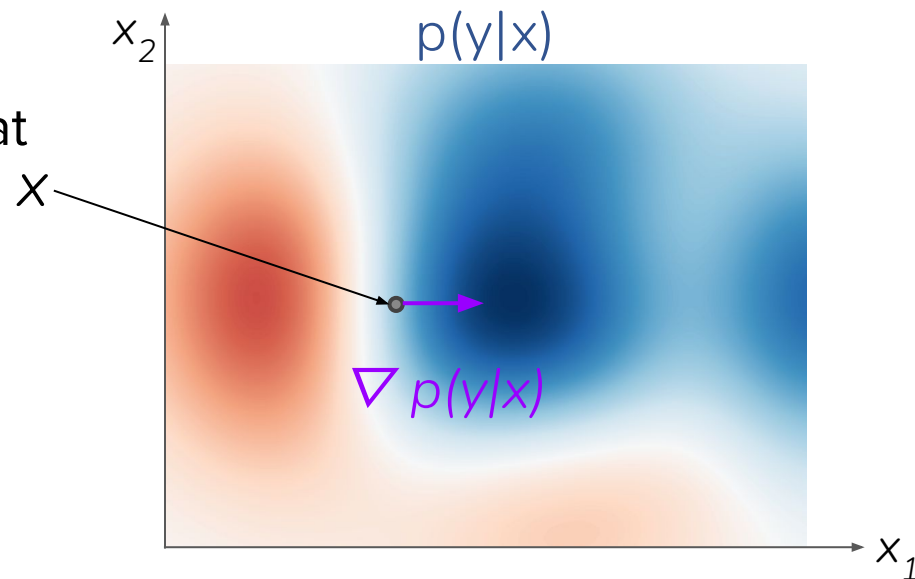
Visualizing the top 5 most important words.

Example of highlighting: Image classification



Highlighting via Input Gradients

- Estimate importance of a feature using derivative of output w.r.t that feature
- i.e., with a “tiny change” to the feature, what happens to the prediction?
- We then visualize the importance values of each feature in a heatmap

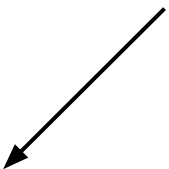


Highlighting via Input Gradients

For NLP, derivative of output w.r.t a feature

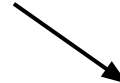
=

derivative of **output** w.r.t an **input token**



What to use as the output?

- Top prediction probability
- Top prediction logits
- Loss (with the top prediction as the ground-truth class)



Token is actually an embedding. How to turn gradient w.r.t embedding into a scalar score?

- Sum it?
- Take an L_p norm?
- Dot product with embedding itself?

Do we normalize values across sentence?

direction lead to
a decrease in
the loss

$$-\nabla_{e(t)} \mathcal{L}_{\hat{y}} \cdot e(t)$$

L1-normalized across all tokens

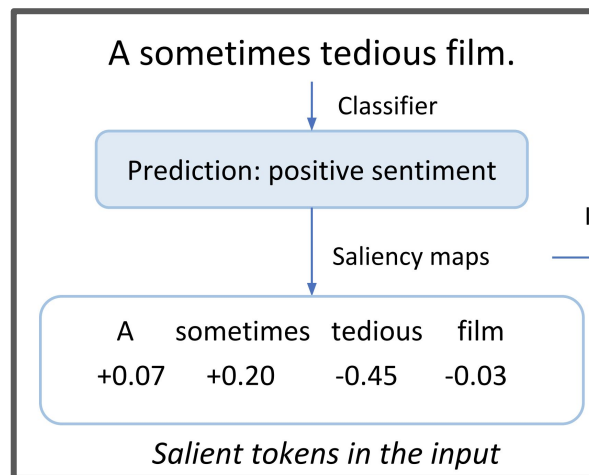
Eqn from [[Han et al. 2020](#)]

Highlighting via Input Gradients

For NLP, derivative of output w.r.t a feature

=

derivative of **output** w.r.t an **input token**



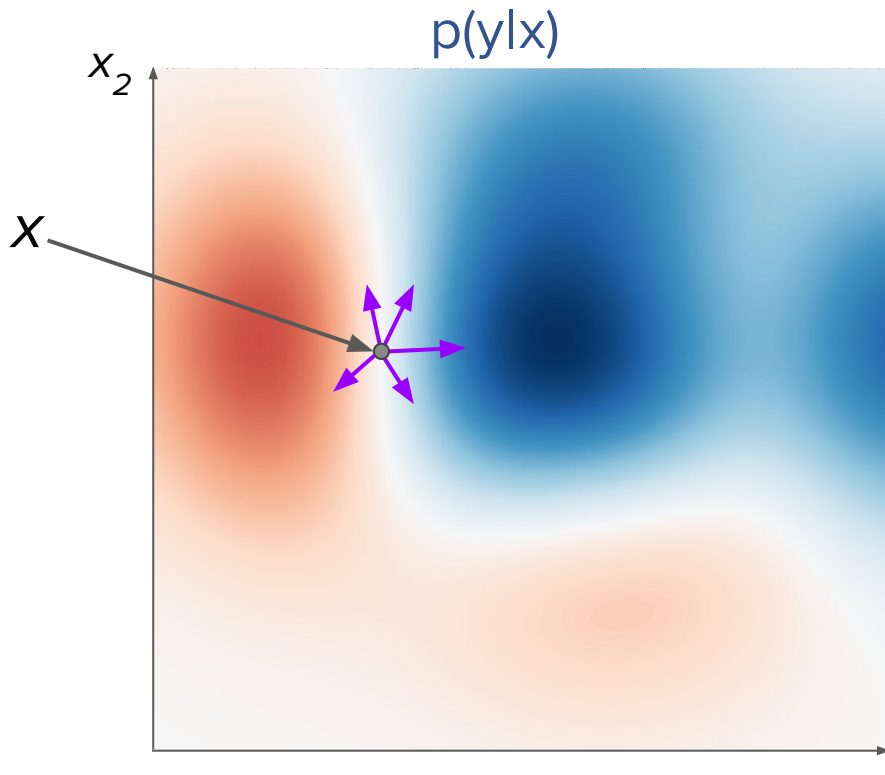
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L1-normalized across all tokens

Problems with Using Gradient for Highlighting

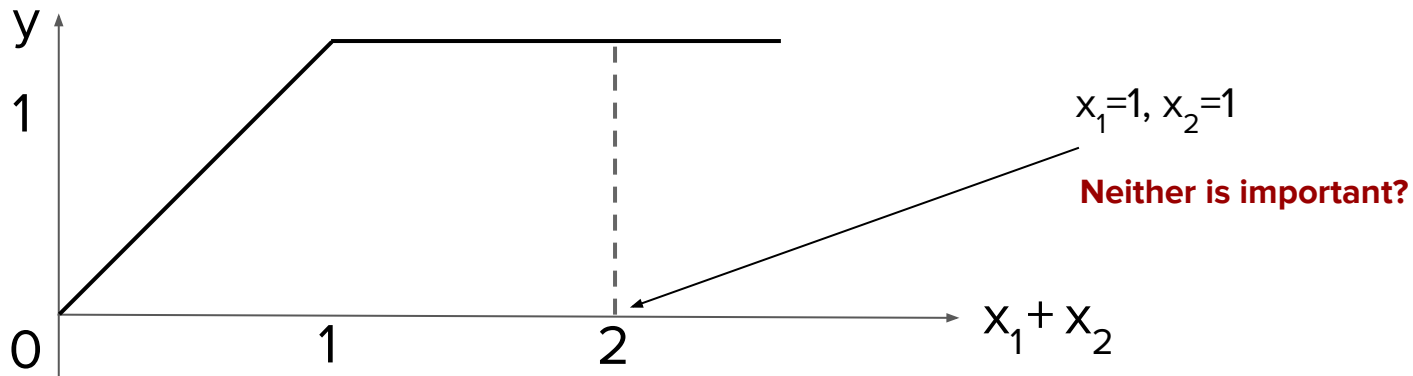
- Too “local” and thus sensitive to slight perturbations



Problems with Using Gradient for Highlighting

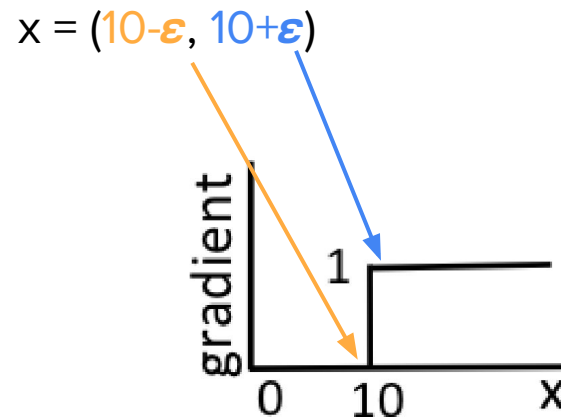
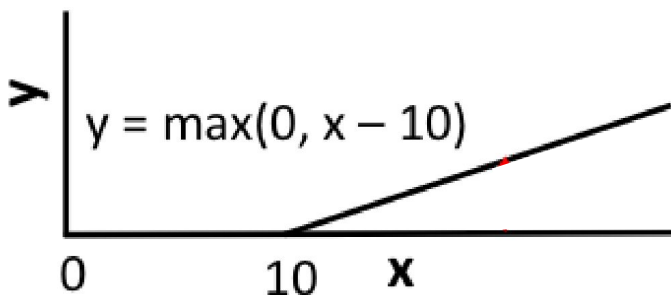
- too “local” and thus sensitive to slight perturbations
- “saturated outputs” lead to unintuitive gradients

$$y = \begin{cases} x_1 + x_2 & \text{when } (x_1 + x_2) < 1 \\ 1 & \text{when } (x_1 + x_2) \geq 1 \end{cases}$$



Problems with Using Gradient for Highlighting

- too “local” and thus sensitive to slight perturbations
- “saturated outputs” lead to unintuitive gradients
- discontinuous gradients (e.g., thresholding) are problematic



Extensions of Vanilla Gradient

- too “local” and thus sensitive to slight perturbations
- “saturated outputs” lead to unintuitive gradients
- discontinuous gradients (e.g., thresholding) are problematic

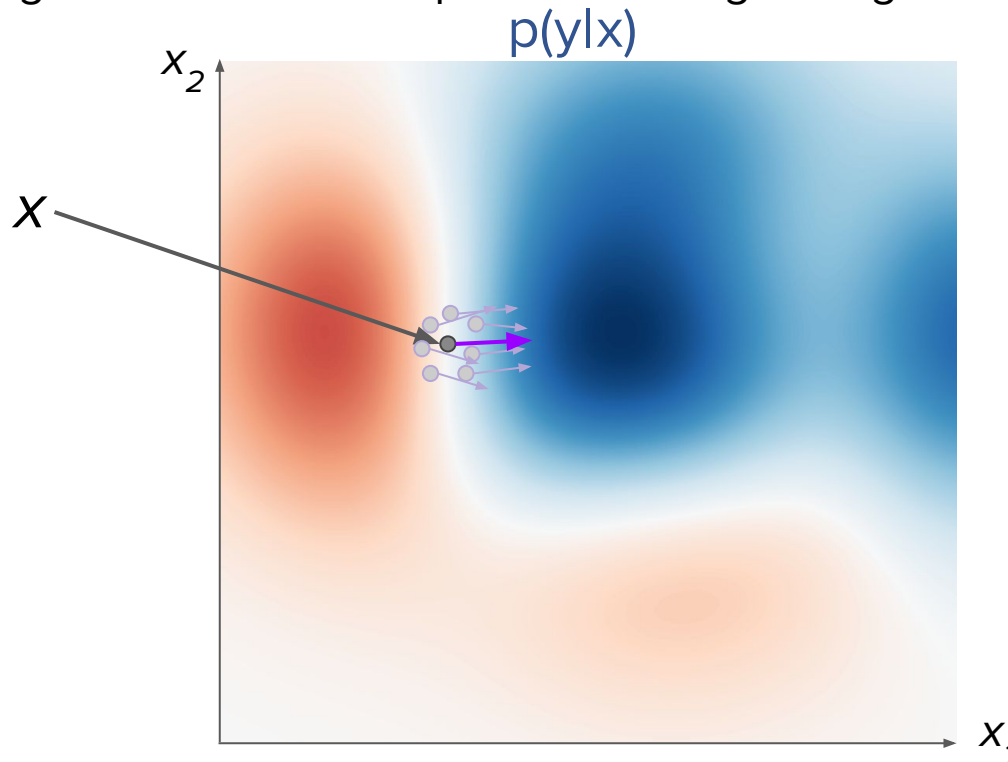
How to mitigate these issues? Don't rely on a single gradient calculation:

- SmoothGrad
- Integrated Gradients

Other approaches, e.g., [LRP](#), [DeepLIFT](#), [GradCAM](#). Not covered here.

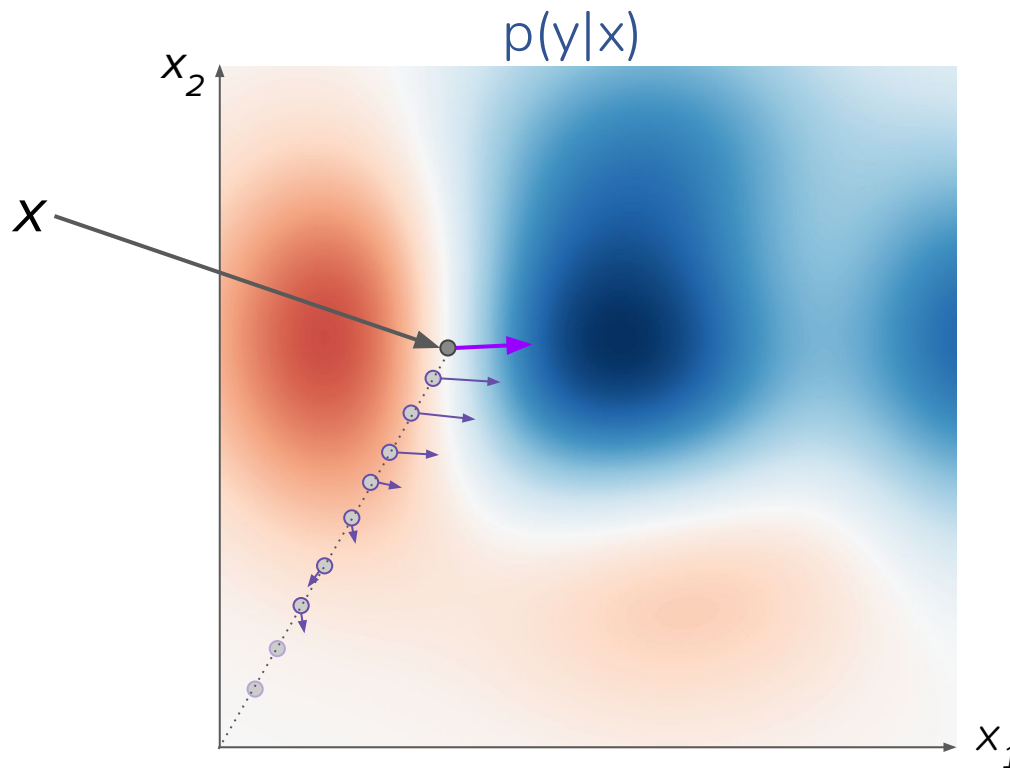
Extensions of Vanilla Gradient

SmoothGrad: add gaussian noise to input and average the gradient



Extensions of Vanilla Gradient

Integrated Gradients: average gradients along path from zero to input



Summary of Gradient-based Highlighting

Positives:

- Fast to compute: single (or a few) calls to backward()
- Visually appealing: spectrum of importance values

Negatives:

- Needs white-box (gradient) access to the model
- Gradients can be unintuitive with saturated or thresholded values
- Difficult to apply to non-classification tasks
- Ignore the interactions between words/pixels (e.g., “not good”)
- Highlighting cannot do anything if a model uses knowledge (such as common sense) that is not explicitly mentioned in the input

Topic 4

Why is this input assigned this answer?



select-then-predict

Can the model inherently assign
an explanation to this answer?

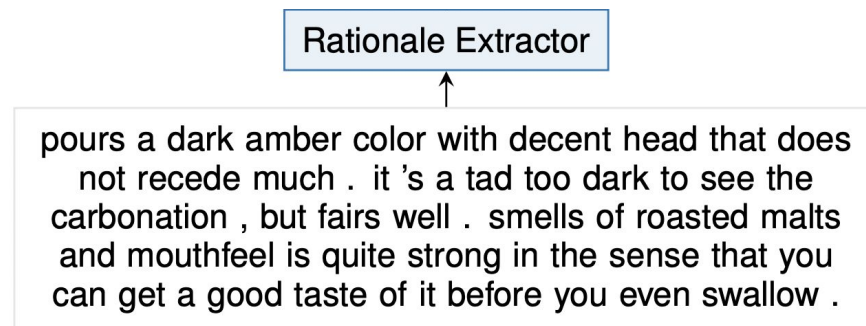
Post-hoc Explainability vs. Inherently Interpretable Models

[Interpretable Machine Learning](#) (book):

- **Intrinsic interpretability** refers to:
 - Models that are considered *inherently interpretable* due to their simple structure, such as *short* decision trees or sparse linear models
 - *Self-explaining* models trained to predict and provide explanations for their predictions; explanations are an integral part of the overall learning problem
- **Post hoc interpretability** refers to the application of interpretation methods after model training

Select-then-Predict

Select-then-Predict



Select-then-Predict

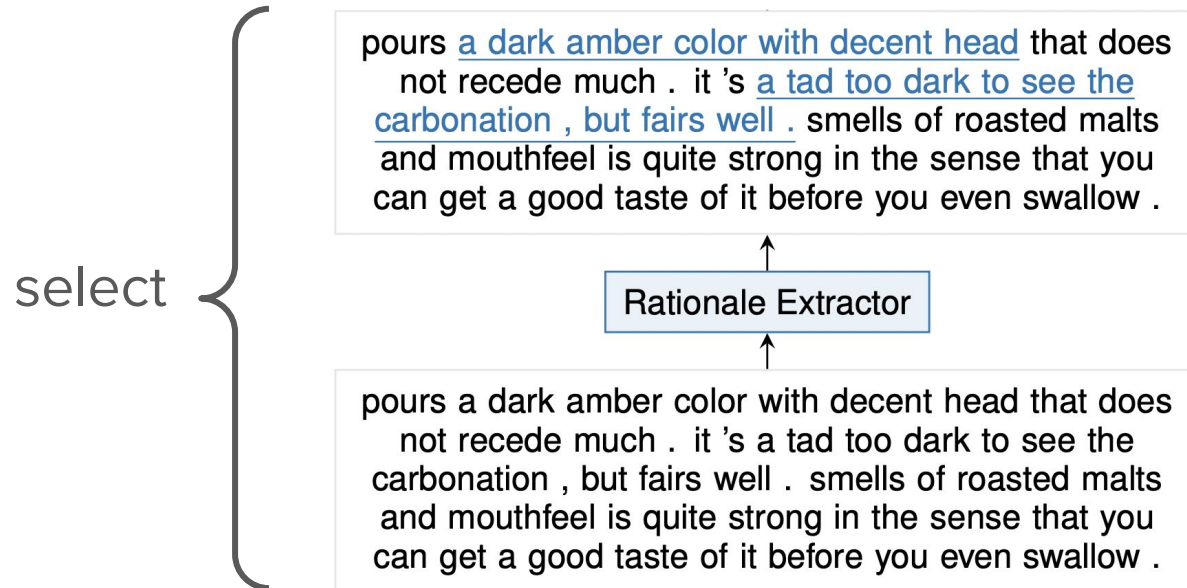
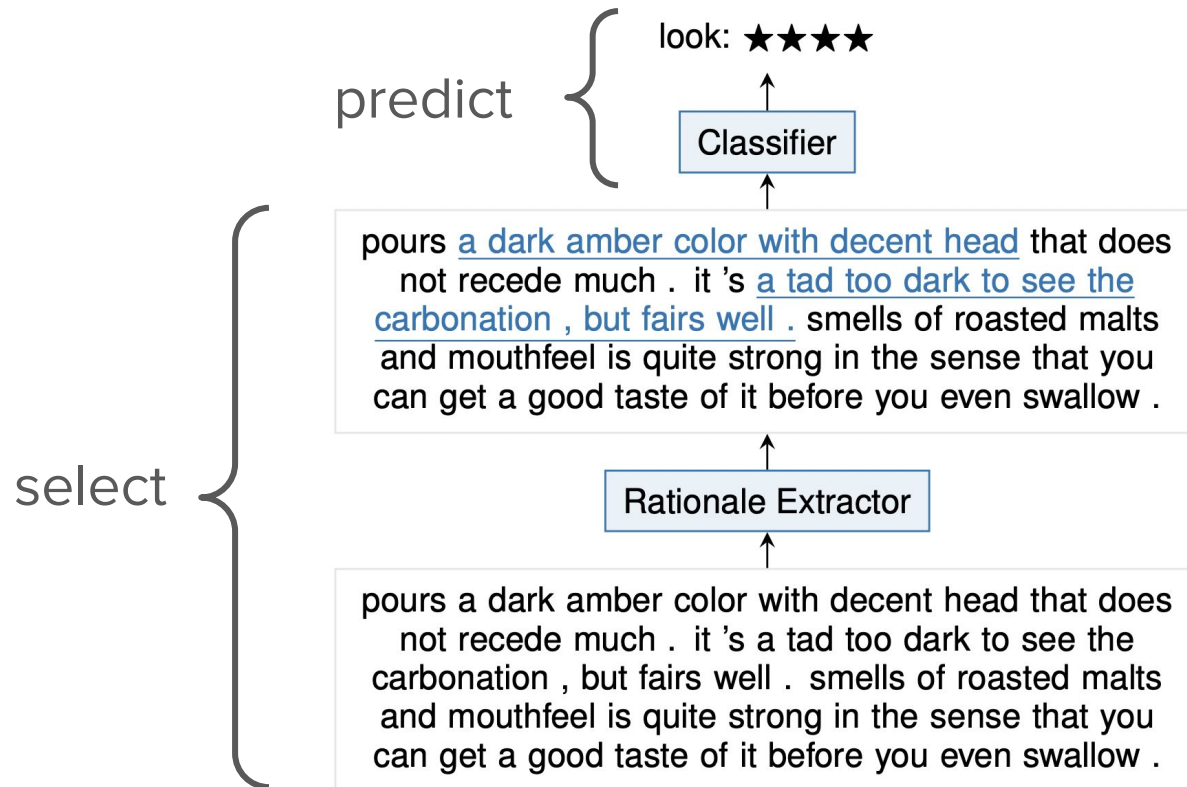


Figure from: [\[Bastings et al., 2019\]](#); Approach proposed in: [\[Lei et al., 2016\]](#)

Select-then-Predict



Highlighting is “an integral part of the overall learning problem”

We do not use human-written highlights to train this model, but weak supervision from the label prediction

Select-then-Predict

Highlights (originally called “rationales”) are defined as subsets of input words that:

1. Represent short & coherent pieces of text
2. Alone suffice for prediction as a substitute of the original text (**sufficiency**)
3. [Yu et al., 2019](#): Represent all the evidence that supports the prediction, i.e., the tokens that are not selected do not predict the label (**comprehensiveness**)

select

pours a dark amber color with decent head that does not recede much . it 's a tad too dark to see the carbonation , but fairs well . smells of roasted malts and mouthfeel is quite strong in the sense that you can get a good taste of it before you even swallow .

Rationale Extractor

pours a dark amber color with decent head that does not recede much . it 's a tad too dark to see the carbonation , but fairs well . smells of roasted malts and mouthfeel is quite strong in the sense that you can get a good taste of it before you even swallow .

Select-then-Predict

Highlights (originally called “rationales”) are defined as subsets of input words that:

1. Represent **short** & **coherent** pieces of text
2. Alone suffice for prediction as a substitute of the original text (**sufficiency**)

Generator:

Assigns a binary value, z_t , to each token in the input: 0=do not highlight, 1=highlight

$$p(\mathbf{z}|\mathbf{x}) = \prod_{t=1}^l p(\mathbf{z}_t|\mathbf{x}, \mathbf{z}_1 \cdots \mathbf{z}_{t-1})$$

$$\Omega(\mathbf{z}) = \lambda_1 \|\mathbf{z}\| + \lambda_2 \sum_t |\mathbf{z}_t - \mathbf{z}_{t-1}|$$

Encoder: Makes the prediction; $\mathbf{y}$ is gold label

$$\mathcal{L}(\mathbf{z}, \mathbf{x}, \mathbf{y}) = \|\text{enc}(\mathbf{z}, \mathbf{x}) - \mathbf{y}\|_2^2$$

$$\text{cost}(\mathbf{z}, \mathbf{x}, \mathbf{y}) = \mathcal{L}(\mathbf{z}, \mathbf{x}, \mathbf{y}) + \Omega(\mathbf{z})$$

Break!



Why is this input assigned this answer?



Instead of training special models,
what explanations can model
internals provide?

Visualizing attention weights became a widely used technique to interpret models' behavior

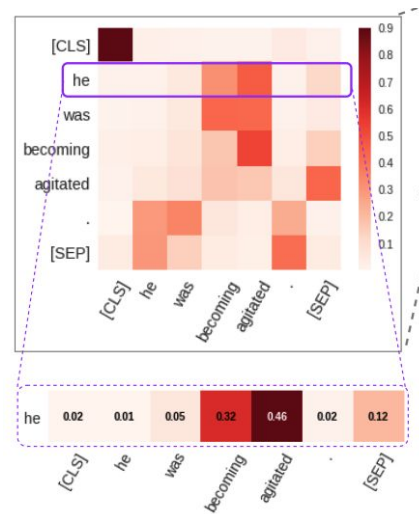
LSTM word attention* [[Bahdanau et al., 2015](#)]

brilliant and moving performances by tom and peter finch

Transformer self-attention

$$\text{softmax}\left(\frac{\begin{matrix} \text{Q} \\ \begin{matrix} \text{6x6 grid} \end{matrix} \end{matrix} \times \begin{matrix} \text{K}^T \\ \begin{matrix} \text{6x6 grid} \end{matrix} \end{matrix}}{\sqrt{d_k}}\right)$$

let's visualize



*[Understanding LSTM Networks](#)

Is Attention Explanation?

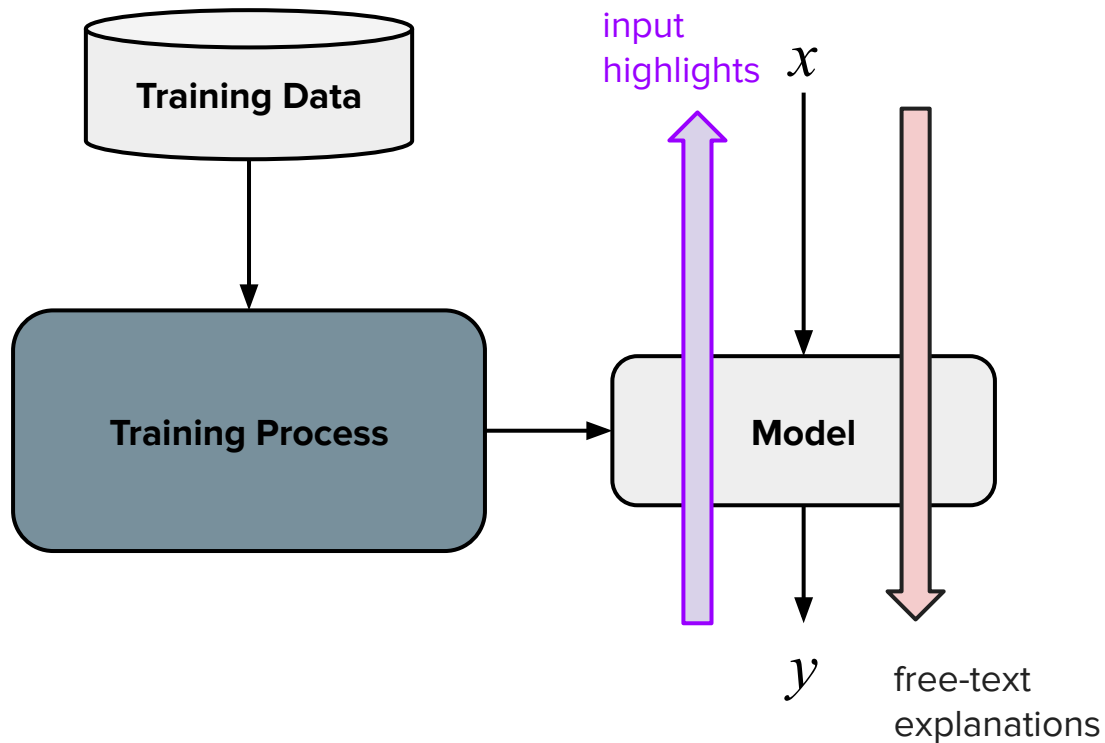
[[Bibal et al., 2022](#)]

But it all started with:

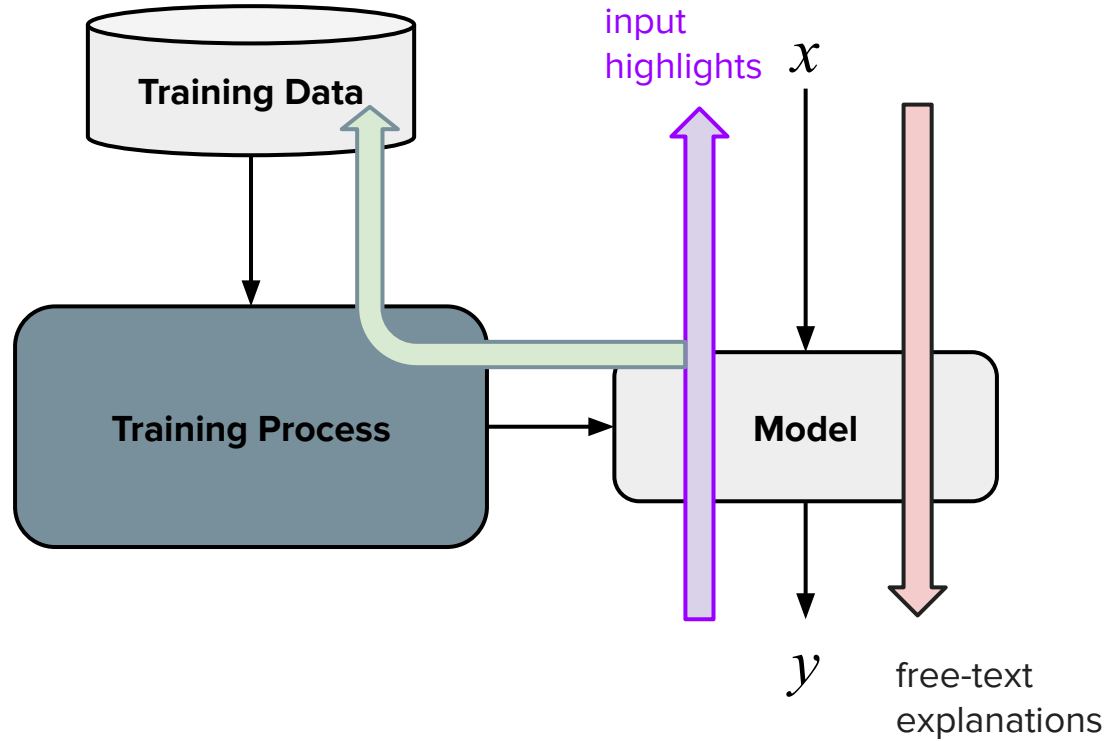
1. Jain and Wallace. [Attention is not Explanation](#). NAACL 2019.
2. Wiegrefe and Pinter. [Attention is not not Explanation](#). EMNLP 2019.

Work	Task	Architecture	Section
Galassi et al. (2020)	NA (survey)	NA (survey)	Section 3
de Santana Correia and Colombini (2021)	NA (survey)	NA (survey)	Section 3
Ras et al. (2021)	NA (survey)	NA (survey)	Section 3
Serrano and Smith (2019)	Topic Classification	HAN	Section 4
Thorne et al. (2019)	Natural Language Inference	LSTM-CRF	Section 4
Mohankumar et al. (2020)	Sentiment Analysis, Text Classification, Natural Language Inference, Paraphrase Detection and Question Answering	LSTM	Sections 4, 8 and 9.1
Ethayarajh and Jurafsky (2021)	NA (theoretical work)	NA (theoretical work)	Section 4
Bai et al. (2021)	Text and Image Classification	CNN	Sections 5 and 9.1
Brunner et al. (2020)	Regression	BERT	Section 5
Sun and Lu (2020)	Text Classification	LSTM	Section 5
Tutek and Šnajder (2020)	Text Classification	LSTM	Sections 5 and 9.1
Clark et al. (2019)	Dependency Parsing and Coreference Resolution	BERT	Section 6
Vig and Belinkov (2019)	Sequence to Sequence	GPT-2	Section 6
Vashishth et al. (2019)	Text Classification, Natural Language Inference, Question Answering and Translation	RNN, Bi-RNN, multi-layer Bi-RNN and HAN	Sections 6 and 8
Neely et al. (2021)	Text Classification and Natural Language Inference	Bi-LSTM and Distil-BERT	Section 7
Ju et al. (2021)	NA (theoretical work)	NA (theoretical work)	Section 7
Liu et al. (2020)	Text Classification	LSTM and BERT	Section 7
Jacovi and Goldberg (2020)	NA (theoretical work)	NA (theoretical work)	Section 7
Sen et al. (2020)	Text Classification	RNN and Bi-RNN	Section 8
Sood et al. (2020)	Question Answering	LSTM, CNN and XLNet	Section 8
Pruthi et al. (2020)	Text Classification	Embedding, Bi-LSTM and BERT	Section 8
Chrysostomou and Aletras (2021)	Text Classification	Bi-LSTM, Bi-GRU, CNN, MLP and BERT	Section 9.1
Moradi et al. (2021)	Translation	LSTM	Section 9.1
Strout et al. (2019)	Text Classification	CNN	Section 9.2
Zhong et al. (2019)	Sentiment Analysis	Bi-LSTM, TreeLSTM, LSTM over SDP and CNN	Section 9.2
Heo et al. (2020)	Classification and Regression	Neural Processes	Section 9.2
Kanchinadam et al. (2020)	Text Classification	LSSVM	Section 9.2
Arous et al. (2021)	Text Classification	SciBERT and AL-BERT	Section 9.2

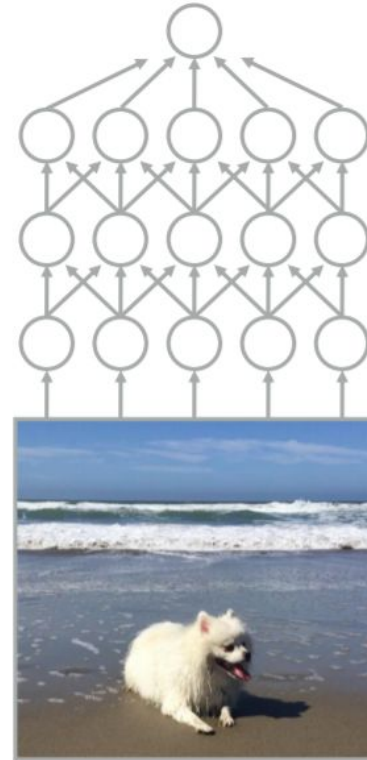
So far...



Data Influence

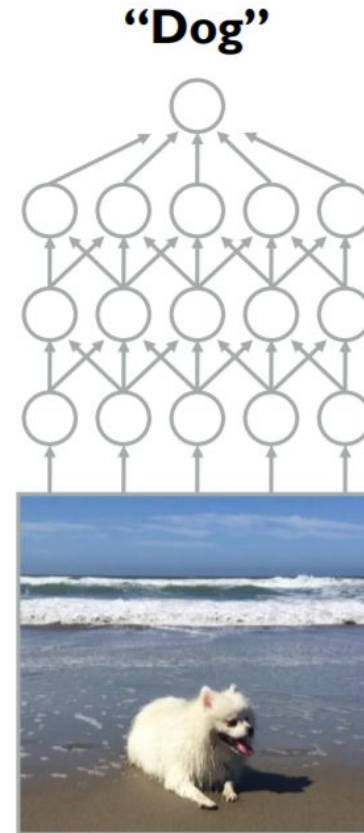


“Dog”





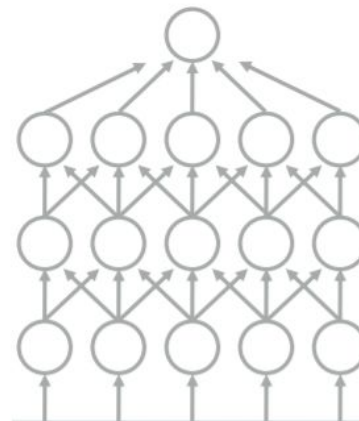
Training





Training

“Dog”



Most
Important

Training data

Why did my model make this prediction?



Data influence

Which training examples were responsible for this prediction?

Why did my model make this prediction?



Data influence

Which training examples were responsible for this prediction?



Influence functions

Which examples, if removed, would change the loss a lot?

Fish



Dog

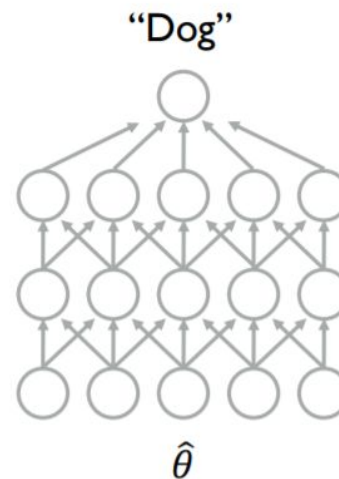


Dog



Training data $z_1, z_2, \dots, z_n$

Pick $\hat{\theta}$ to minimize $\frac{1}{n} \sum_{i=1}^n L(z_i, \theta)$

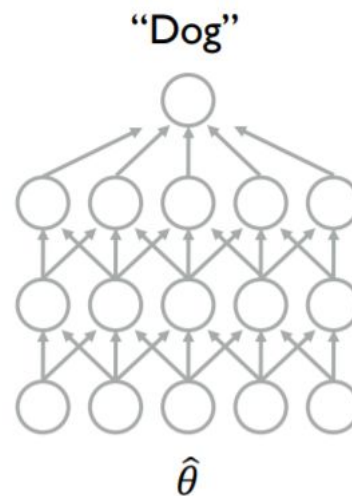




Training data $z_1, z_2, \dots, z_n$

z_{train}

Pick $\hat{\theta}$ to minimize $\frac{1}{n} \sum_{i=1}^n L(z_i, \theta)$



Slide credit: [Pang Wei Koh](#)



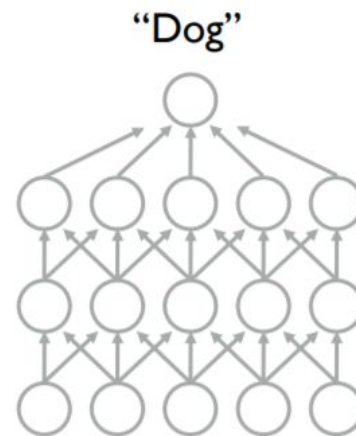
Training data $z_1, z_2, \dots, z_n$

z_{train}

Pick $\hat{\theta}$ to minimize $\frac{1}{n} \sum_{i=1}^n L(z_i, \theta)$

Pick $\hat{\theta}_{z_{train}}$ to minimize

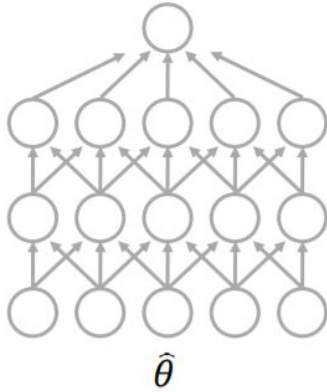
$$\frac{1}{n} \sum_{i=1}^n L(z_i, \theta) - \frac{1}{n} L(z_{train}, \theta)$$



$\hat{\theta}_{z_{train}}$

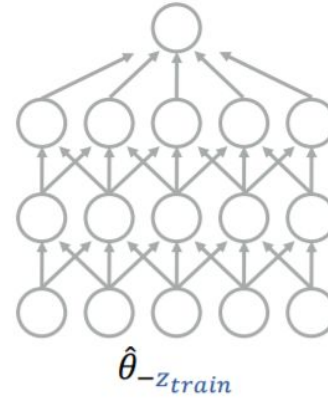
Slide credit: [Pang Wei Koh](#)

“Dog” (82% confidence)



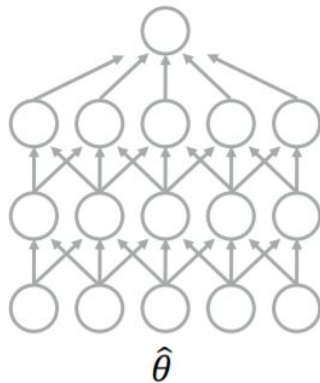
vs.

“Dog” (79% confidence)



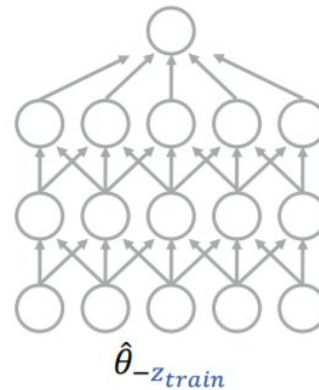
Test input z_{test}

“Dog” (82% confidence)



vs.

“Dog” (79% confidence)



What is $L(z_{test}, \hat{\theta}_{-z_{train}}) - L(z_{test}, \hat{\theta})$?

Problem

Repeatedly removing a training point and retraining the model is too slow

Solution

Approximation via influence functions
(a classical technique from the 1970s)

Influence Functions

- Goal: Compute $L(\mathbf{z}_{test}, \hat{\theta}_{-\mathbf{z}_{train}}) - L(\mathbf{z}_{test}, \hat{\theta})$

$$\hat{\theta}_{-\mathbf{z}_{train}} \stackrel{\text{def}}{=} \operatorname{argmin}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(z_i, \theta) - \frac{1}{n} L(\mathbf{z}_{train}, \theta)$$

- Idea:
 - Assume $\frac{1}{n}$ is small
 - Use calculus to compute effect of removing ϵ weight from $\mathbf{z}_{train}$
 - Linearly extrapolate to removing $\frac{1}{n}$ weight

[[Koh and Liang, 2017](#)]

Slide credit: [Pang Wei Koh](#)

Influence Functions

- Goal: Compute $L(\mathbf{z}_{test}, \hat{\theta}_{-\mathbf{z}_{train}}) - L(\mathbf{z}_{test}, \hat{\theta})$

$$\hat{\theta}_{-\mathbf{z}_{train}} \stackrel{\text{def}}{=} \operatorname{argmin}_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n L(\mathbf{z}_i, \theta) - \frac{1}{n} L(\mathbf{z}_{train}, \theta)$$

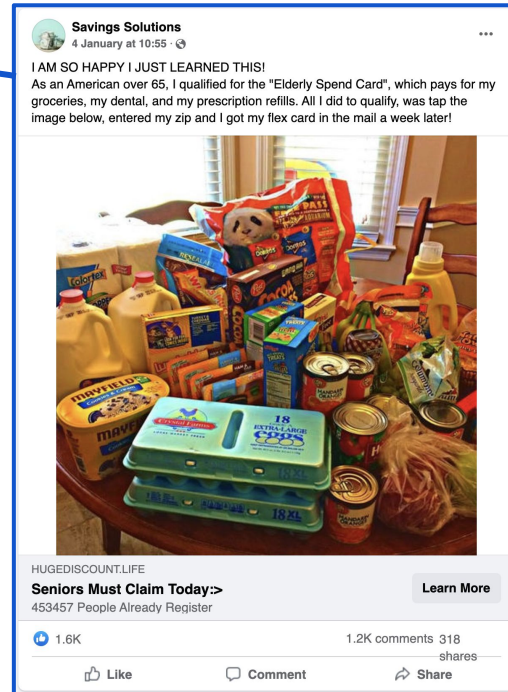
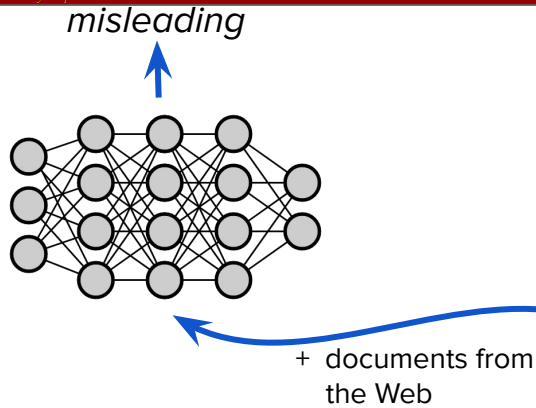
This classic result from the statistics tells us how to calculate influence of **upweighting** a data point on the parameters

Why did my model make this prediction?



Contrastive Editing

What would have happened if I changed
the input to my model?



*Why is my post misleading?
How can I change it to
make it clear/correct?*

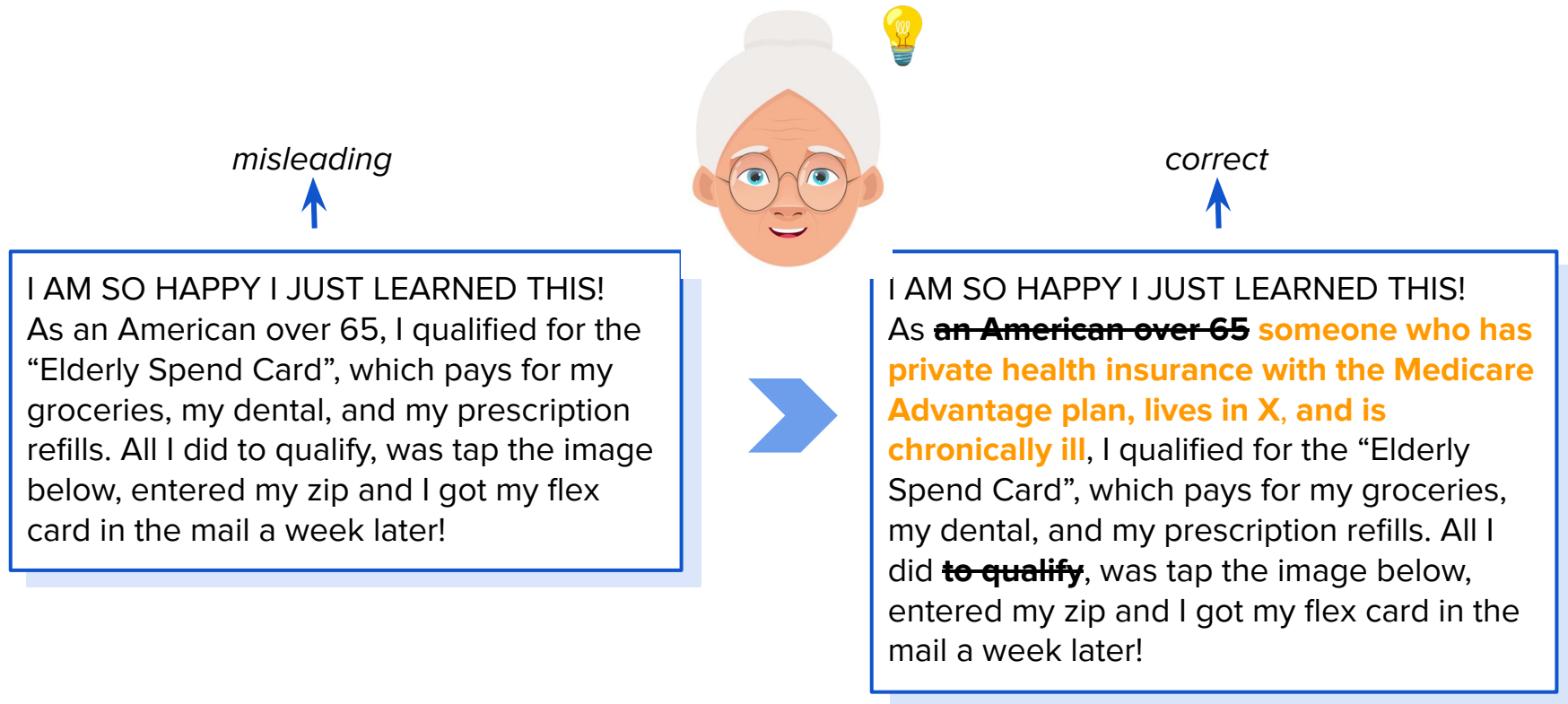
misleading



*Why is my post misleading?
How can I change it to
make it clear/correct?*

I AM SO HAPPY I JUST LEARNED THIS!
As an American over 65, I qualified for the
“Elderly Spend Card”, which pays for my
groceries, my dental, and my prescription
refills. All I did to qualify, was tap the image
below, entered my zip and I got my flex
card in the mail a week later!

Contrastive explanations: explain **how to minimally modify**
the input to change the prediction to something else



Contrastive explanations: explain **how to minimally modify**
the input to change the prediction to something else

Contrastive Explanations via **Contrastive Editing**

The key idea:

“*Why P not Q?*” $\Rightarrow$ “How to change the answer from P to Q?”
 $\Rightarrow$ By making a **contrastive minimal edit**

A minimal edit to the input that causes the model output to change to the contrast case
has hallmark characteristics of a human contrastive explanation:

- $\rightarrow$ cites contrastive features
- $\rightarrow$ selects a few relevant causes

Contrastive Explanations via **Contrastive Editing**

Question:

Ann and her children are going to Linda's home _____.

(a) by bus (b) by car (c) on foot (d) by train

Why **“by train”** (d) and not **“on foot”** (c)?

How to change the answer from **“by train”** (d) to **“on foot”** (c)?

Context:

...Dear Ann, I hope that you and your children will be here in two weeks. My husband and I will go to meet you at the train station. Our town is small...

MICE-Edited Context:

...Dear Ann, I hope that you and your children will be here in two weeks. My husband and I will go to meet you at ~~the train station~~ **your home on foot**. Our ~~town~~ **house** is small...

Ross, Marasović, Peters (2021)

Explaining NLP Models via Minimal Contrastive Editing (MiCE)



Goal:

Explain a **Predictor** model by *automatically* finding a **minimal edit** to the input that **causes the model output to change to the contrast case**

A very high-level idea of 🐭:

- Use an **Editor** model to keep masking input words & filling masked positions until an edit that changes the label predicted by **Predictor** is found
- Simultaneously, minimize the masking percentage, i.e., the edit size

input: Sylvester Stallone has made some crap films in his lifetime, but this has got to be one of the worst. A totally dull story...

the contrast label (foil)



label: positive **input:** Sylvester Stallone has made some crap films in his lifetime, but this has got to be one of the worst. A totally dull story...

the contrast label (foil)

label: positive input: Sylvester Stallone has made some crap films in his lifetime, but this has got to be one of the worst. A totally dull story...



mask $n\%$ of input tokens

label: positive input: Sylvester Stallone has made some <mask> films in his lifetime, but this has got to be one of the <mask>. A totally <mask> story...

the contrast label (foil)

label: positive input: Sylvester Stallone has made some crap films in his lifetime, but this has got to be one of the worst. A totally dull story...



mask $n\%$ of input tokens

label: positive input: Sylvester Stallone has made some <mask> films in his lifetime, but this has got to be one of the <mask>. A totally <mask> story...



sample 15 spans at each masked position

1. label: positive input: Sylvester Stallone has made some **good** films in his lifetime, but this has got to be one of the **worst**. A totally **novel** story...
2. label: positive input: Sylvester Stallone has made some **great** films in his lifetime, but this has got to be one of the **greatest of all time**. A totally **boring** story...
- ...
15. label: positive input: Sylvester Stallone has made some **wonderful** films in his lifetime, but this has got to be one of the **greatest**. A totally **tedious** story...

the contrast label (foil)

label: positive input: Sylvester Stallone has made some crap films in his lifetime, but this has got to be one of the worst. A totally dull story...

mask *n*% of input tokens

label: positive input: Sylvester Stallone has made some <mask> films in his lifetime, but this has got to be one of the <mask>. A totally <mask> story...

sample 15 spans at each masked position

1. label: positive input: Sylvester Stallone has made some **good** films in his lifetime, but this has got to be one of the **worst**. A totally **novel** story...
2. label: positive input: Sylvester Stallone has made some **great** films in his lifetime, but this has got to be one of the **greatest of all time**. A totally **boring** story...
- ...
15. label: positive input: Sylvester Stallone has made some **wonderful** films in his lifetime, but this has got to be one of the **greatest**. A totally **tedious** story...

get the logit of the contrast label

$$l(pos) = 0.2$$

$$l(pos) = 0.6$$

$$l(pos) = 0.65$$

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample 15 spans at masked positions

×

**4 different values of n
to minimize the edit**



4*15=60 samples

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample 15 spans at masked positions

×

**4 different values of n
to minimize the edit**



$4 * 15 = 60$ samples



rank 60 samples w.r.t. the logit of the contrast label

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample 15 spans at masked positions

×

**4 different values of n
to minimize the edit**



$4 * 15 = 60$ samples



rank 60 samples w.r.t. the logit of the contrast label



keep top-3 samples **beam**

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample 15 spans at masked positions

×

**4 different values of n
to minimize the edit**



$4 * 15 = 60$ samples



rank 60 samples w.r.t. the logit of the contrast label



keep top-3 samples **beam**

**if a contrastive
edit is found**

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample **15** spans at masked positions

× **4 different values of n to minimize the edit**

Repeat for every instance in the beam at most **2 more rounds**

↓
4*15=60 samples

↓
rank **60** samples w.r.t. the logit of the contrast label

↓
keep top-**3** samples **beam**

1. Prepend the contrast label to the input
2. Mask $n\%$ of the input tokens
3. Sample 15 spans at masked positions

×

**4 different values of n
to minimize the edit**

How to pick masking positions?

Based on token importance for the original prediction

Rank input tokens based on the gradient magnitude of the model we're explaining

Mask top- $n\%$ of **ranked** tokens

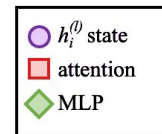
Why did my model make this prediction?



Activation Patching

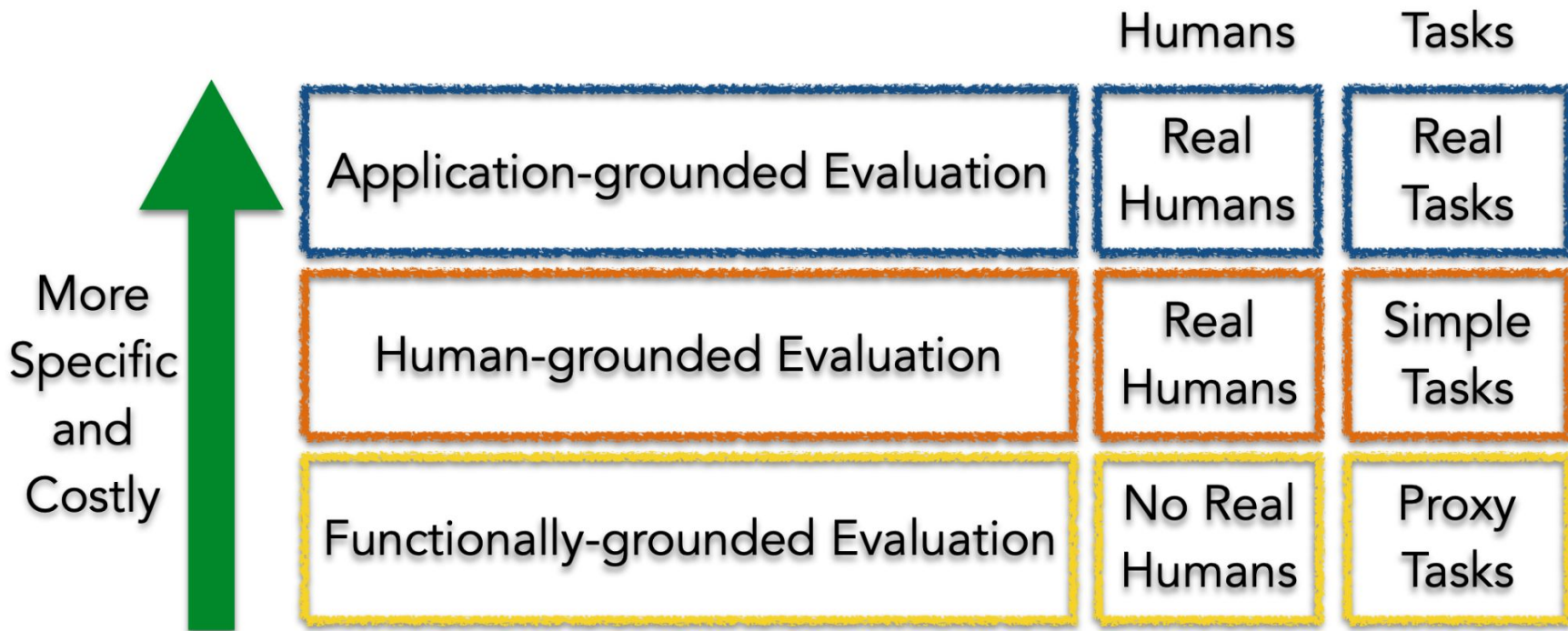
What internal mechanisms inform a prediction?

Activation patching for localizing knowledge



What makes a good explanation?

Doshi-Velez & Kim's Taxonomy of Evaluation



Doshi-Velez & Kim's Taxonomy of Evaluation

Functionality-Grounded Evaluation: No “real humans”; Proxy tasks

Some formal definition of interpretability as a proxy for explanation quality asserted by a MLer

Appropriate when a method is not yet mature to be evaluated with human subjects

Challenges:

- Determine what proxies to use
- Determine an optimization strategy

Examples:

- Show that an inherently-interpretable model performs as good as non-interpretable alternative
- Show that one method gives a sparser solution than the other

Doshi-Velez & Kim's Taxonomy of Evaluation

Human-Grounded Evaluation: Real humans; Proxy tasks

Simpler human-subject experiments that target the quality of an explanation, but not their usefulness for specific, realistic end-goals

Closer to the what we should evaluate to say whether explanations are beneficial but allow to recruit a bigger subject pool and are less expensive

Examples:

- Overlap between automated and human-authored explanations?
- Human judgments of whether an explanation justifies an answer
- A person (or a model; these days GPT-4, although not without issues: see [Singhal et al., 2023](#)) chooses one of the two explanations that they find of higher quality
- Forward or counterfactual simulation

Forward Simulatability

It measures the ability of a person (or another model) to simulate the outcome of the AI on an instance level

$\text{Model}(\text{Input}) \rightarrow \text{Label}^{(m)}$

$\text{ExplainabilityMethod}(\text{Model}, \text{Input}, \text{Label}^{(m)}) \rightarrow \text{Explanation}$

$\text{Human}(\text{Input}, \text{Explanation}) \rightarrow \text{Label}^{(h)}$

Forward simulatability measures:

$\text{Label}^{(m)} = \text{Label}^{(h)}$

Counterfactual Simulatability

It approximates a person's understanding of model behavior by assessing whether the person can change the model

$\text{Model}(\text{Input}^{(\text{orig})}) \rightarrow \text{Label}_1^{(m)}$

$\text{ExplainabilityMethod}(\text{Model}, \text{Input}^{(\text{orig})}, \text{Label}_1^{(m)}) \rightarrow \text{Explanation}$

$\text{Human}(\text{Input}^{(\text{orig})}, \text{Label}_1^{(m)}, \text{Explanation}) \rightarrow \text{Input}^{(\text{new})}$

$\text{Model}(\text{Input}^{(\text{new})}) \rightarrow \text{Label}_2^{(m)}$

Counterfactual simulatability measures:

$\text{Label}_2^{(m)} = y_2$

Where y_2 is any other label that the evaluator set that is not $\text{Label}_1^{(m)}$

Doshi-Velez & Kim's Taxonomy of Evaluation

Application-Grounded Evaluation: Real humans; Real tasks

Human experiments within a real application:

*If the researcher has a concrete application in mind—such as working with doctors on diagnosing patients with a particular disease—the **best way to show that the model works is to evaluate it with respect to the task**: doctors performing diagnoses.*

It directly tests the objective that the system is built for

⇒ performance w.r.t. that objective gives strong evidence of success

Why are application-grounded evaluations scarce? 🤔

NLP explainability has unique technical challenges:

- Features are a sequence of high-dimensional non-interpretable vectors
- An arbitrary number of features
- Continuous representations of discrete inputs
- Explaining models with billions of parameters
- Pretrained models
- Inherently interpretable models (linear models, short decision trees) performing nowhere close to LLMs



So, prior NLP explainability work has mostly focused on overcoming these challenges

Many realistic applications of have become evident/possible only with recent advances

There is a prevailing perspective that this now needs to change since explainability methods passed proof-of-concept tests

What can you do in this space?

- [How To Become A Mechanistic Interpretability Researcher](#) → Great resource to get started in this space!
- Loads of great mech interp researchers don't actually have a PhD!



Program ▾

Blog

Announcing Goodfire's Fellowship Program for Interpretability Research

PUBLISHED

October 9 2025

Launch your career in AI alignment & security

The MATS Program is an independent research and educational seminar program that connects talented researchers with top mentors in the fields of AI alignment, transparency, and security. The program runs for 12 weeks with in-person cohorts in Berkeley and London, where MATS fellows conduct research while attending talks, workshops, and networking events with other members of the AI research community. Top-performing fellows can extend their impactful research for an additional 6 months with continued funding, mentorship, and community support.

Learn more: **Summer 2026**

Anthropic Fellows Program for AI safety research: applications open for May & July 2026

"This is an exceptional opportunity to join AI safety research, collaborating with leading researchers on one of the world's most pressing problems." — Jan Leike

This Lecture

A brief primer on explaining model decisions, mostly taken from research work in this community in the past ~5 years!

- A brief history of how (and why) we think about explanations
- A short primer on different cutting edge methods for explaining model decisions:
 - Uncertainty Estimation
 - Natural Language Explanations
 - Highlighting spans
 - Inherently Interpretable models
 - Token Attribution
 - Training Data
 - Contrastive Editing
 - Modern Interpretability: Activation Patching
- How can we evaluate explanations?

Thank you!

